Atomic Structure

1. Introduction

1.1 Defination of Atom

An atom is the smallest unit of an element that retains its chemical properties. It consists of a nucleous containing protons and neutrons, surrounded by electrons in various energy levels.

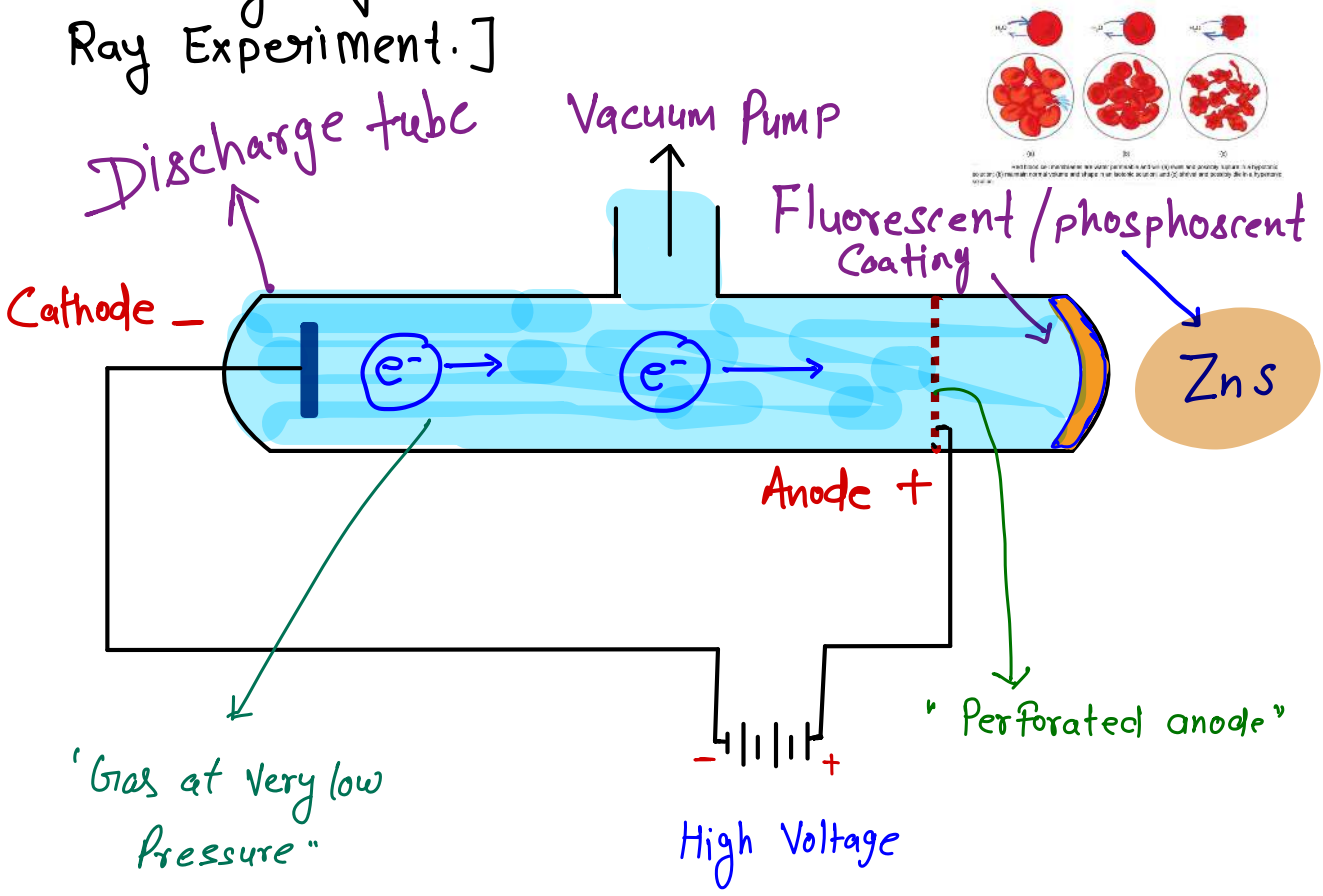
1.2 Subatomic Particles:-

atoms are composed of three Fundamental particles

- Electrons (e^-): Negatively charged mass = $9.1093837 \times 10^{-31}$ kg.
- Protons (p^+): Positively charged mass = $1.67262192 \times 10^{-27}$ kg.
- Neutron (n^0): Neutral charge, mass = $1.674927 49 \times 10^{-27}$ kg.

2. Discovery of Subatomic Particles:

2.1 Discovery of Electron [J.J. Thomson's Cathode Ray Experiment.]

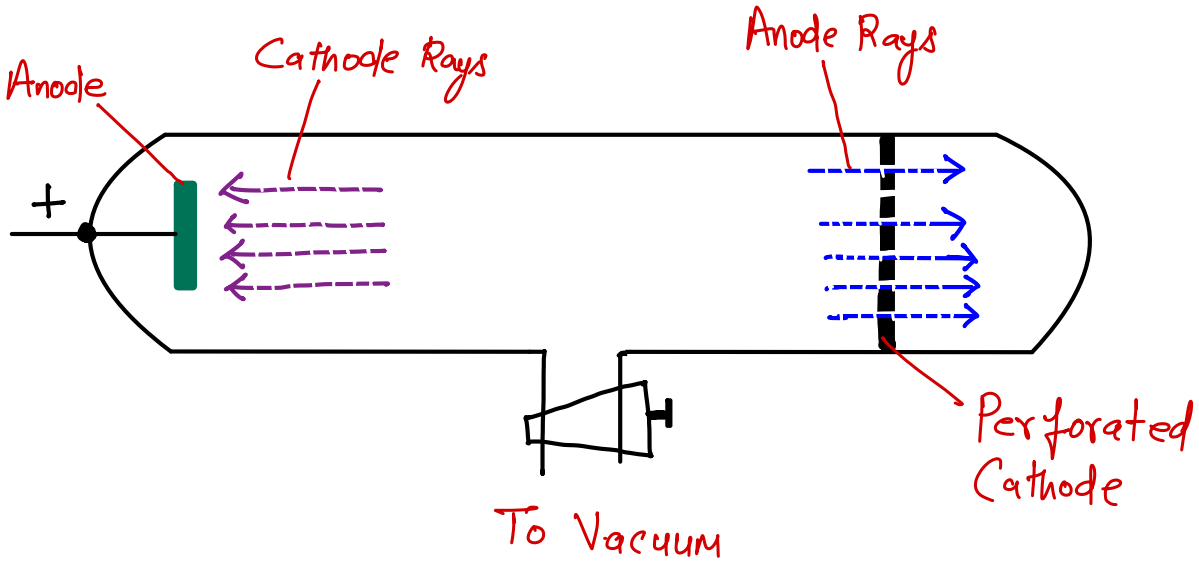


J.J. Thomson discovered electrons using the

Cathode Ray Tube Experiment:

- A high Voltage was applied to gases at low Pressure, Producing Cathode rays.
- These rays were deflected by electric and mangnetic fields, Proving they were negatively charged.
- The charged-to-mass ratio (e/m) of an electron was determined as C/kg.

2.2 Discovery of Proton (Goldstein's Canal Ray Experiment)

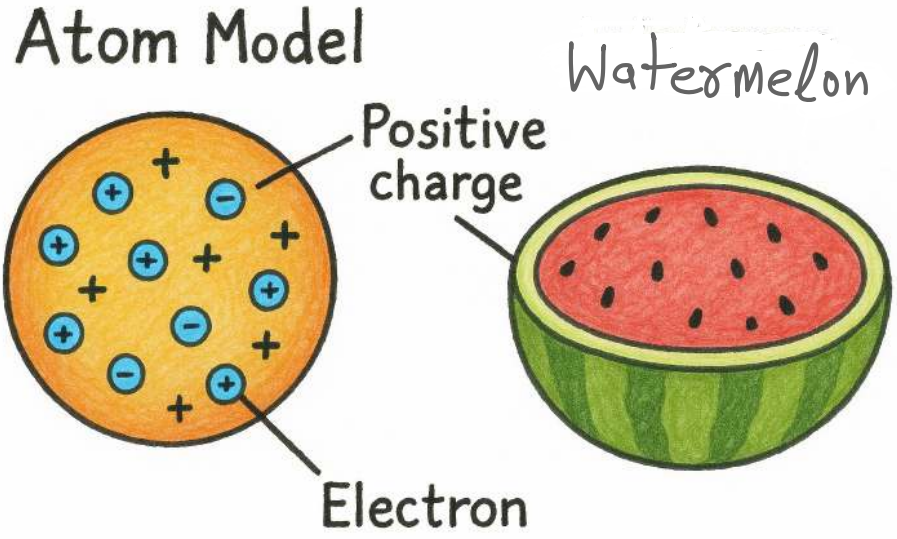


E. Goldstein discovered protons using the Canal Ray Experiment:

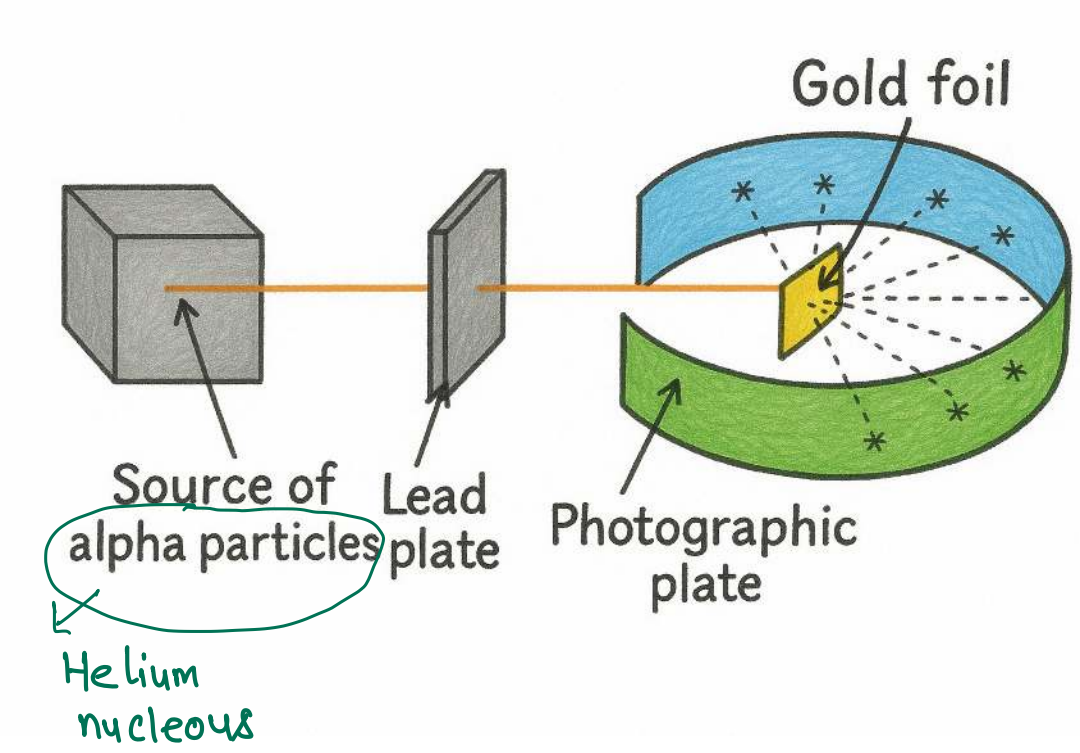
- When high Voltage was applied to gases, Positively charged rays (Canal Rays) were observed moving in the opposite direction of Cathode rays.
- These rays were Composed of Protons.

3 → Atomic Models

3.1 → Thomson's atomic Model



- Proposed by J.J. Thomson in 1897, this model is also known as the Plum Pudding Model or Watermelon Model.
- According to this model, an atom consists of a positively charged sphere in which negatively charged electrons are embedded like plums in a pudding or seeds in a watermelon.
- The positive charged is uniformed spread throughout the atom, and the negatively charged electrons balance it to maintain overall neutrality.
- This model was later discarded as it failed to explain the nucleus and atomic structure, which was later clarified by Rutherford's Model.



- Conducted by Ernest Rutherford in 1911, this Experiment aimed to study the structure of the atom by bombarding a thin goid foil with alpha Particles (helium nuclei)

- The observations of the experiment were:-

  - Most alpha particles passes through the foil without defilection, indicated that atoms are mostly empty space.

  - Some Particles were deflected at small angles. Suggesting the Presence of a Concentrated Positive charge.

- A Few Particles rebounded at Large angles, leading to the Conclusion that a dense and Positively charged nucleus exists at the Centre of the atom.

- This experiment disproved Thomson's Plum Pudding Model and led to the development of Rutherford's Nuclear Model of the Atom. which stated that electrons orbit a dense, positively charged nucleus.

4→ Properties of Fundamental Particles.

Name	Symbol	Absolute charge (C)	Relative charge	Mass (Kg)	Mass (u)	App. Mass (u)
Electron	e	$-1.602176 \times 10^{-19}$	-1	9.109382×10^{-31}	0.00054	0
Proton	p	$+1.602176 \times 10^{-19}$	+1	$1.6726216 \times 10^{-27}$	1.00727	1
Neutron	n	0	0	1.674927×10^{-27}	1.00867	1

5. Dual Nature of Light

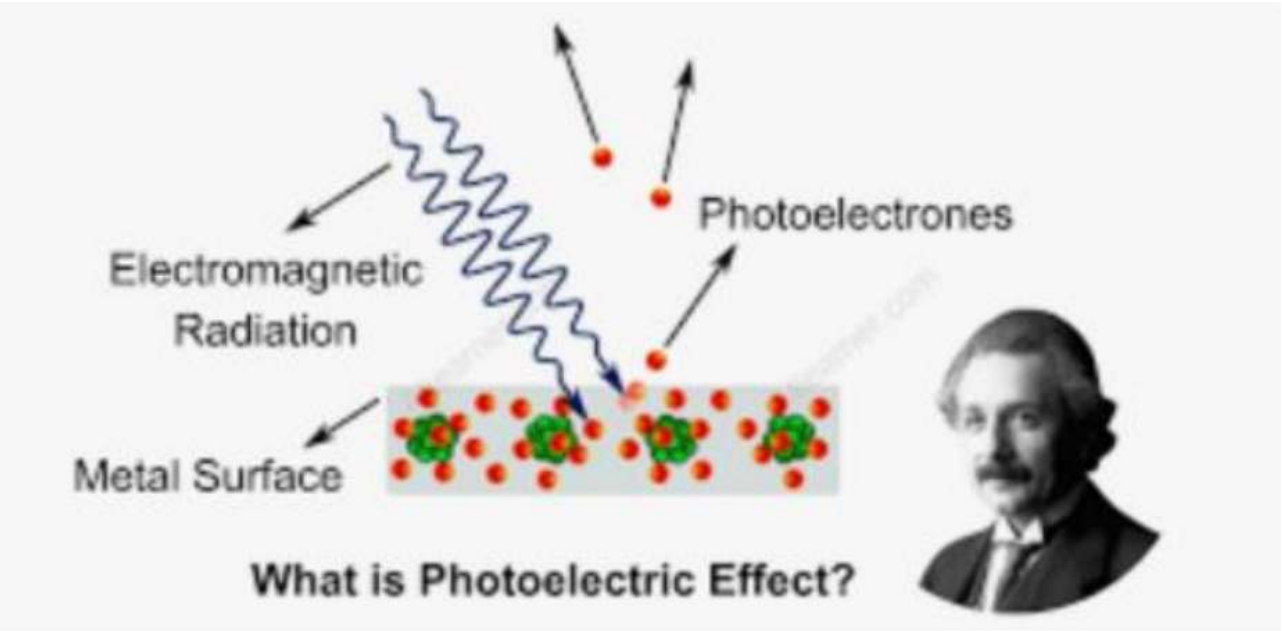
Light exhibits both wave-like and Particle-like behaviour, depending on the experiment being Performed. This Concept is known as the dual....

Nature of light and was established through various experiments.

- Wave Nature of light
  - Demonstrated by Young's Double-Slit Experiment, which showed interference and diffraction, proving that light behaves as a wave.
  - Explained using Maxwell's electromagnetic wave theory, which describes light as oscillating electric and magnetic Fields.
- Particle Nature of Light
  - Confirmed by the photoelectric effect (Einstein, 1905), where light ejects electrons from a metal Surface.
  - Introduced the Concept of photons, which are discreate packet (quonta) of energy.
    - Einstein's equation $E = h\nu$
      h = plank's Constant
      ν = Frequency of Light

Property	Wave Nature	Particle Nature
Phenomenon	Interference Diffraction	Photoelectric Effect. Compton effect
Explained by	Maxwell's Electro-magnetic Theory	Quantum Mechanics (Planck, Einstein)
Key Concept	Continuous propagation as waves	Light Consists of discreate photons
Energy Relation	Depends on intensity	$E = h\nu$ (depends on frequency)

Photoelectric effect



The photoelectric effect is the phenomenon where electrons are ejected from a metal surface when light of a certain frequency shines on it. This experiment provided strong evidence for the particle nature of light and led to the devolopment of quantum mechanics.

key observations

1- **Threshold Frequency**: Electrons are only emitted if the light's frequency is above a certain minimum value (threshold frequency, ν_0)

2. **Instantaneous Emission**: Electrons are ejected immediately after light falls on the surface, proving that energy is not accumulated over time.

3. **Intensity vs. Energy**: Increasing the intensity of light increases the number of ejected electrons but does not increase their energy.

4. **Frequency vs. Kinetic Energy**: The kinetic energy of emitted electrons depends on the frequency of light, not its intensity.

Einstein explained this using quantum theory:

$$h\nu = h\nu_0 + KE_{max}$$

where:

h = Planck's Constant

ν = Frequency of incident light

ν_0 = Threshold Frequency

$K.E_{max}$ = Maximum kinetic energy of emitted electrons.

This equation showed that light Consists of photons, each carrying energy $E = h\nu$, proving the particle nature of light.

7 → Bohr's Theory

Niels Bohr proposed a model of the atom in 1913 to explain the stability of atoms and the discrete line spectra of elements. His theory is based on quantum mechanics and Modefied Rutherford's atomic Model.

7.1 Postulates of Bohr's Atomic Model

1. Electrons move in Fixed orbits (energy levels) around the nucleus without radiating energy.

2. Only Certain discrete orbits are allowed, where the angular momentum of an electron is quantized.

$$mvr = nh/2\pi$$

Where $n = 1, 2, 3, \ldots\ldots$ (Principal quantum number)

h = Planck's Constant
m = Mass of electron
v = Velocity of electron
and r = radius of orbit.

3. Energy is absorbed or emitted when an electron jumps from one orbit to another, given by:

$$\Delta E = h\nu$$

where ν = Frequency of emitted / absorbed radiation.

4. The radius of an electron's orbit is given by:

$$r = 0.529\ n^2/Z\text{A}^\circ$$

Where ε_0 = permittivity of Free Space
e = charge of electron.

5 The energy of an electron in an orbit is given by

$$E_n = -13.6\frac{Z^2}{n^2}$$

This explains why electrons in lower orbits have more negative energy.

7.2 Limitations of Bohr's Theory

- Could not explain the spectra of atoms with multiple electrons.
- Failed to explain Fine spectral lines and Zeeman effect.
- Did not Consider wave-particle duality of electrons.

This model laid the foundation for quantum mechanics and was later refined by Schrodinger's wave equation.

8. Hydrogen Spectrum

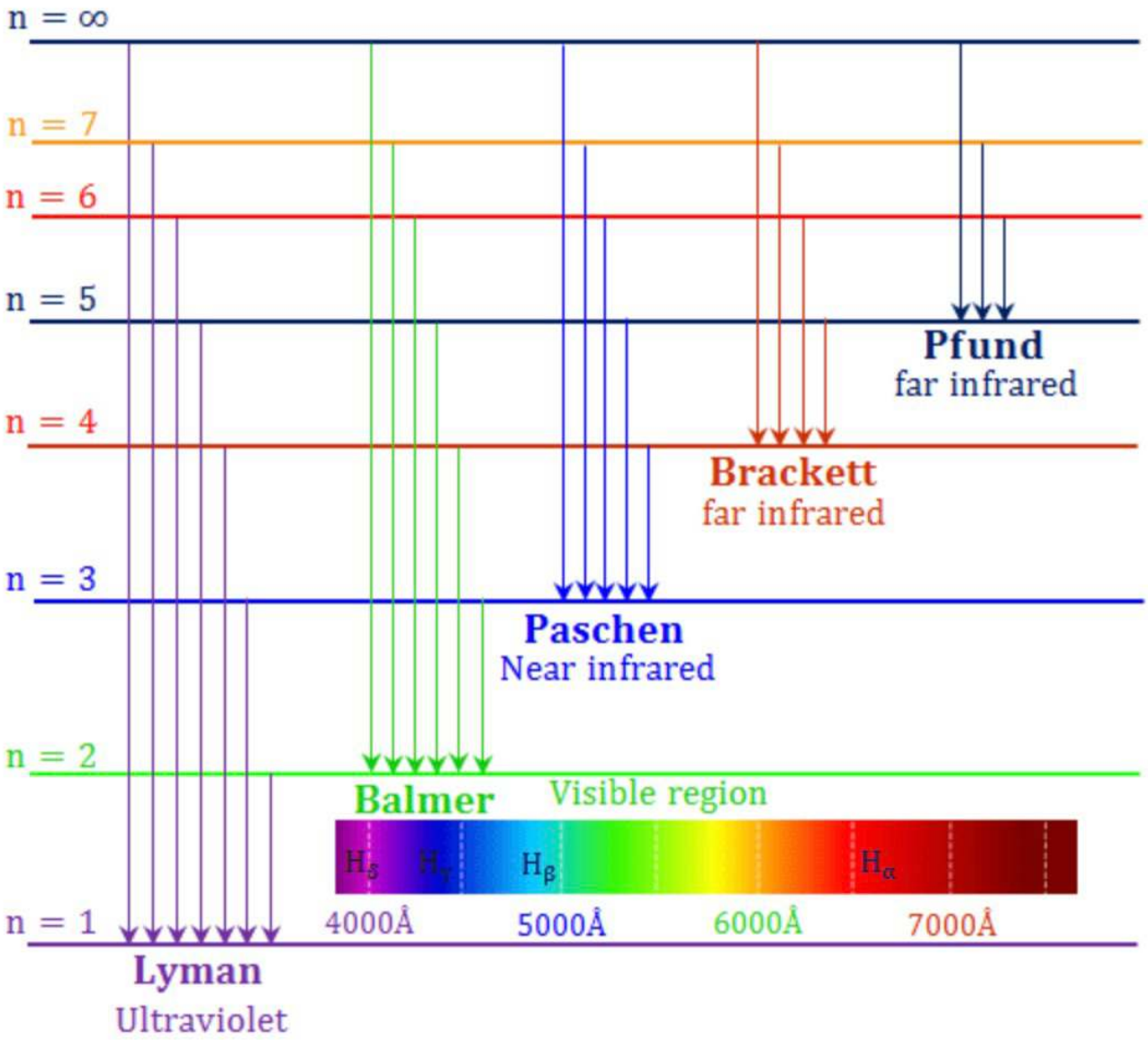
The hydrogen Spectrum Consists of discrete spectral lines that appear when hydrogen atoms absorb or emit energy. This Spectrum is an important proof of quantized energy levels in atoms.

Formation of Hydrogen Spectrum

- When an electron in a hydrogen atom absorbs energy, it moves to a higher energy level (excited state)

- When it returns to a lower energy level, it emits a photon of a specific wavelength.
- These emitted photons form the line spectrum of hydrogen.

8.1 Spectral Series of Hydrogen



- The hydrogen Spectrum is divided into different Series based on the Final energy level of the electron transition:.

Series	Region	Final Orbit (n_2)	Formula (1/λ)
Lyman	Ultraviolet	$n_2 = 1$	$\frac{1}{\lambda} = R_H\left(\frac{1}{1^2} - \frac{1}{n_1^2}\right)$
Balmer	Visible	$n_2 = 2$	$\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{n_1^2}\right)$
Paschen	Infrared	$n_2 = 3$	$\frac{1}{\lambda} = R_H\left(\frac{1}{3^2} - \frac{1}{n_1^2}\right)$
Brackett	Infrared	$n_2 = 4$	$\frac{1}{\lambda} = R_H\left(\frac{1}{4^2} - \frac{1}{n_1^2}\right)$
Pfund	Infrared	$n_2 = 5$	$\frac{1}{\lambda} = R_H\left(\frac{1}{5^2} - \frac{1}{n_1^2}\right)$

here R_H is Ryberg Constant

9. De Broglie's Wavelength

Louis de Broglie proposed the wave-particle duality of matter, stating that Particles like electrons also exhibit wave-like behaviour. This led to the Concept of matter waves.

De Broglie's Wavelength Formula

The wavelength (λ) of a moving particle is given by:

$$\lambda = h/mv$$

Where:

h = Planck's Constant (6.626×10^{-34} Js)
m = mass of the Particle
v = Velocity of the Particle

9.1 Significance of De Broglie's Hypothesis

1. Explains the stability of electron orbits: Electrons exist in specific orbits because only whole-number wavelengths fit into them.

2. Led to quantum mechanics: Laid the foundation for Schrodinger's wave equation.

3. Verified by Davisson-Germer Experiment: Confirmed that electron exhibit diffraction, proving their wave nature.

10 Heisenberg's Uncertainty Principle

Werner Heisenberg proposed the Uncertainty Principle, which states that it is impossible to simultaneously determine the exact position and momentum of a particle with absolute precision.

Mathematical Expression

$$\Delta x \cdot \Delta p \geq h/4\pi$$

where

Δx = uncertainty in position

$\Delta p = m\Delta v$ = uncertainty in Momentum

h = Planck's Constant 6.626×10^{-34} J s

11. Quantum Mechanical Model of the Atom

The Quantum Mechanical Model, developed by Erwin Schrodinger, describes electrons in an atom using wave functions (ψ) and probability distributions instead of Fixed orbits (as in Bohr's Model)

Key Features

1 - Electron Clouds & Orbitals:

- Electrons are found in regions Called orbitals not fixed Circular paths.
- Orbitals are defined by quantum numbers.

2. Wave Function (ψ) & Probability:

- Schroginger's equation determines the wave function ψ, which describes the probability of finding an electron in a given space.
- The probability density is given by $(\psi)^2$.

3. Quantum Number's:

- Principle Quantum Number (n) → Size of orbital
- Azimuthal Quantum Number (l) → shape of orbital (s, p, d, f)
- Magnetic Quantum Number (m_l) → Orientation of orbital.
- Spin Quantum Number (m_s) → Electron spin ($+\frac{1}{2}$ or $-\frac{1}{2}$)

Significance

- Explains multi-electron atoms more accurately that Bohr's model.
- Accounts for Heisenberg's Uncertainty Principle
- Forms the basis of Modern atomic theory and quantum chemistry.

12. Quantum Numbers

Quantum Numbers describe the position, energy and behaviour of electrons in an atom. There are four quantum numbers, each defining a specific property of an electron:

1. Principle Quantum Number (n)

- Represents the energy level (or shell) of an electron.
- Possible values: $n = 1, 2, 3, 4, \ldots$
- Larger n means higher energy and greater distance from the nucleus.

2. Azimuthal Quantum Number (l)

Defines the shape of the orbital.

Value $l = 0$ to $n-1$

Corresponding orbitals:

- $l = 0 \longrightarrow$ s - orbital (spherical)
  $l = 1 \longrightarrow$ p - orbital (dumbbell)
  $l = 2 \longrightarrow$ d - orbital (cloverleaf)
  $l = 3 \longrightarrow$ F - orbital (Complex shape)

3. Magnetic Quantum Number (m_ℓ)

- Determines the orientation of an orbital in space
- Values: $m_\ell = -l$ to $+l$ (including 0)

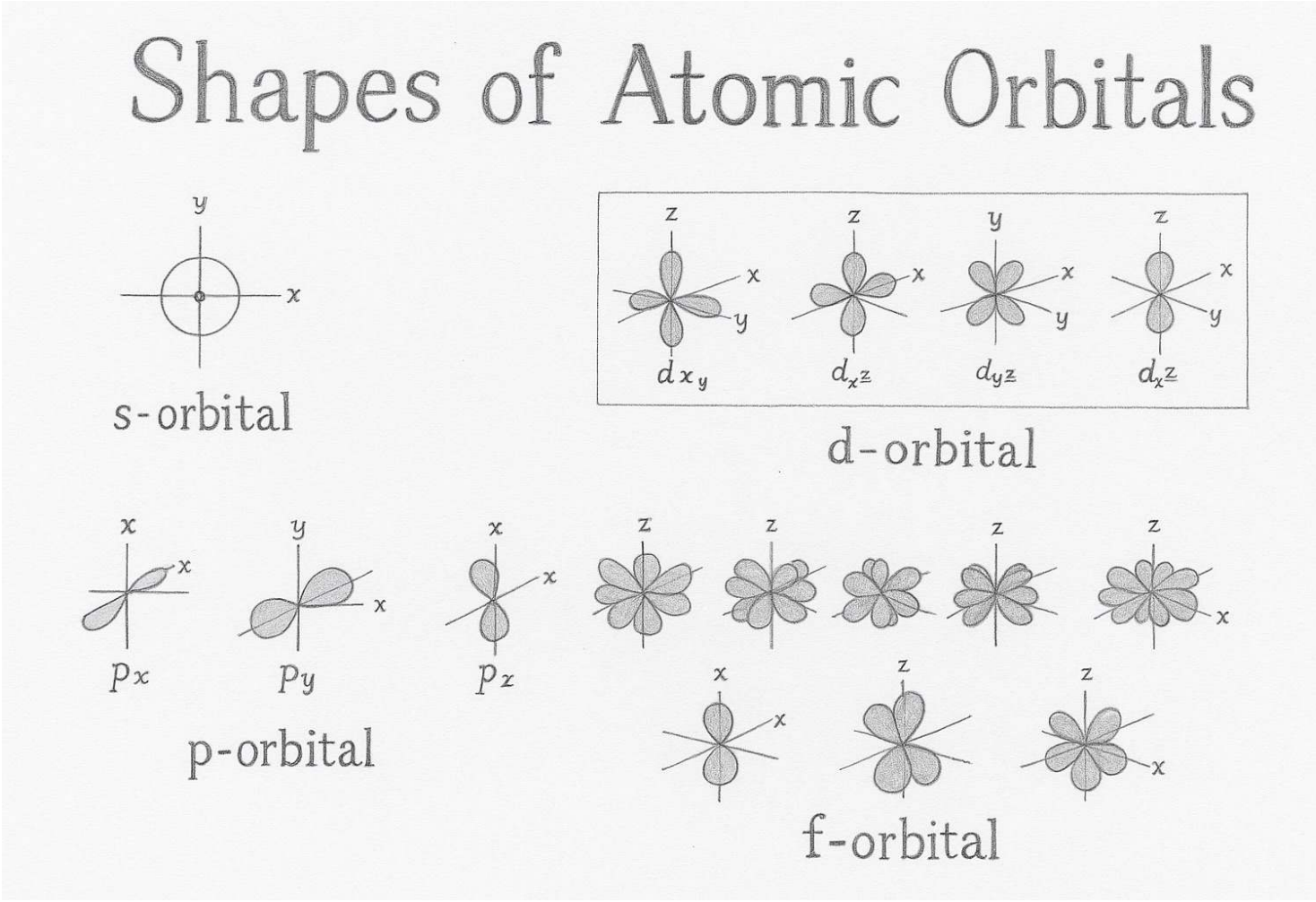
4 Spin Quantum Number (m_s)

- Defines the spin of an electron.
- Possible values: $+\frac{1}{2}$ (clockwise spin) or $-\frac{1}{2}$ (Counter clockwise spin)

Significance of Quantum Numbers

- Explain electronic Configuration and chemical bonding
- Determine the Periodic table structure and element properties.
- Used in Spectroscopy and quantum mechanics.

13 - Shape of Orbitals:.



Orbitals are regions around the nucleus where the probability of finding an electron is maximum. The shape of an orbital is determined by the azimuthal quantum number (ℓ)

1. s orbital ($\ell = 0$)

Shape: Spherical

Example: 1s, 2s, 3s

characteristics:

- Symmetrical around the nucleus.
- No directional preference.
- Increase in size with increasing Principal quantum number (n)

2- p-orbital ($l = 1$)

Shape: Dumbbell - Shaped

Example: 2P, 3P, 4P

characteristics:

- Three orientations: P_x, P_y, P_z (along x, y and Z axes)
- Found in energy levels $n \geq 2$
- Node at the nucleous (Zero probability of finding an electron)

3- d-orbital ($l = 2$)

Shape: cloverleaf (except one orbital which is dumbbell)

Example: 3d, 4d, 5d

characteristics:

- Five orientations: d_{xy}, d_{xz}, d_{yz}, $d_{x^2-y^2}$, d_{z^2}

- Found in energy levels $n \geq 3$
- More Complex shapes, influencing transition metal chemistry.

4. f - Orbital ($l = 3$)

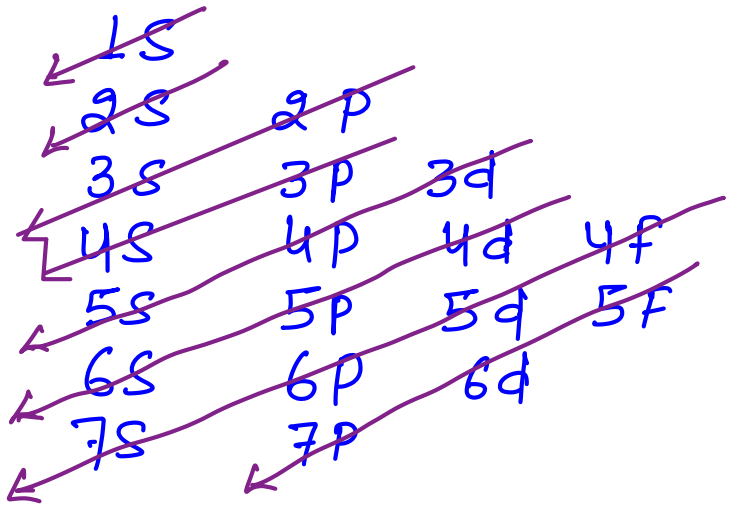
Shape: Highly Complex, multiple lobes.

Example: 4F, 5F

Characteristics:

- Seven orientations.
- Found in energy levels $n \geq 4$
- Important for Lanthanides and actinides chemistry.

14. Aufbau's Principle



The Aufbau Principle states that electrons fill atomic orbitals in order of increasing energy level, following the lowest energy first Rule!.

Order of filling Orbitals :-

- Electrons occupy orbitals in the Sequence:

$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s \rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d \rightarrow 7p$

15. Hund's Rule of maximum multiplicity :-

Hund's Rule states that electrons fill degenerate (equal-energy) orbitals singly first before pairing up. This ensures maximum multiplicity, leading to a more stable electronic Configuration.

Key Point :-

1. Electrons occupy empty orbitals first in the Same Sub level (p, d, f) before pairing.
2. All singly occupied orbitals have parallel spins to minimize repulsion.
3. Results in maximum number of unpaired electrons, increasing stability.

16. Pauli's Exclusion Principle

Pauli's Exclusion Principle states that no two electrons in an atom can have the same set of all four quantum numbers (n, l, m_l, s)

Key Point:-

1. Each orbital can hold a maximum of two electrons with opposite spins ($+\frac{1}{2}, -\frac{1}{2}$)

2. Ensures uniquenes in the arrangement of electrons in an atom.

3. Explains the structure of electron shells and subshells.

17. Stability of Completely Filled and Half-Filled orbitals.

Certain electronic configurations are more stable when orbitals are either completely filled or half-filled due to the following reasons:-

1. Symmetrical Distribution of electrons:-

- Completely filled and half-filled orbitals have a more symmetrical arrangement, leading to lower energy and higher stability.

2. Exchange Energy

- Electrons with parallel spins in degenerate orbitals can exchange positions which lowers the energy.

- More exchange possibilities in half-filled and fully filled orbitals result in extra stability.